PERT TIME ESTIMATES

three guesses, one time!

Operations Management and Strategic Management (Operations Management)

IN SIMPLE WORDS When activity time is uncertain, PERT takes three guesses and finds one weighted average time.

1 OPTIMISTIC (A), PESSIMISTIC (B)

a is the shortest time if all goes well. b is the longest time if almost everything goes wrong.

a is the minimum, b is the maximum

2 MOST LIKELY TIME (M)

The time the activity usually takes under normal conditions. It is the most probable estimate.

m gets the heavy weight of 4

1. PERT is used when activity times are uncertain, such as in research or new projects.
2. Expected time (te) is a weighted average of a, m and b, using beta distribution weights.
3. Variance shows the uncertainty of an activity: a wider gap between b and a means higher variance.
4. Critical path is found using te values; project variance uses critical activities only.



Swipe for the formula →

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THE PERT FORMULAS

PERT Three Time Estimates

learn all three, score full

Expected time of an activity

$$TE = \frac{a + 4m + b}{6}$$

te = expected time; a = optimistic; m = most likely; b = pessimistic

Variance of an activity

$$\sigma^2 = \frac{(b - a)^2}{36}$$

σ^2 = variance; b – a = range of estimates; 36 = 6 squared

Probability of completion (Z value)

$$Z = \frac{Ts - Te}{\sqrt{(\Sigma\sigma^2 \text{ of critical path})}}$$

Ts = scheduled time; Te = expected project time (critical path total); $\Sigma\sigma^2$ = sum of critical activity variances

VARIANCE ADDS, SD DOES NOT! Add the variances of critical activities, then take the square root to get the project standard deviation.

Swipe for a memory trick →

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STUDY TRICK

PERT Three Time Estimates

1-4-1 over 6

ONE-FOUR-ONE, SIX IS DONE

Weights are 1 for a, 4 for m, 1 for b. They total 6, so divide by 6. Variance uses the gap (b − a) over 6, squared.

HOW TO ATTEMPT IT

1 Find te for every activity → 2 Draw network, find critical path →

3 Add te on critical path = Te → 4 Find σ^2 of critical activities, add →

5 Compute Z, read probability

EXAM TIP

In MCQs, first check whether the question asks for te, variance or standard deviation, then pick the matching formula.

DOUBT BUSTER

Why is m multiplied by 4 and not a or b? The most likely time is assumed to be the most probable value, so it gets more weight. a and b are rare extremes and get weight 1 each.

Swipe for common mistakes →

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COMMON MISTAKES

PERT Three Time Estimates

avoid these slips!

✗ Using (a + m + b) / 3 for expected time

✓ PERT uses te = (a + 4m + b) / 6. The simple average ignores the weights.

✗ Writing variance as (b − a) / 6

✓ That is the standard deviation. Variance = [(b − a) / 6]².

✗ Adding variances of all activities for the project

✓ Add variances of critical path activities only.

✗ Adding standard deviations of critical activities

✓ Add the variances first, then take the square root for project standard deviation.

TOMORROW'S CMA INTER MCQ TESTS THIS! Revise it tonight, then try the MCQ.

Save this for revision!

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